How Could a Complex Eye Evolve Gradually?

Javalab Evolution of the Eye · Javalab

Year 10	~30 minutes	Science · Biology

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Simulation: https://javalab.org/en/evolution_of_the_eye_en/
Curriculum links: AC9S10U02 · evolution

Learning intentions

- I can describe the intermediate stages in the evolution of a camera-type eye.
- I can explain the selective advantage offered by each incremental change.
- I can evaluate the claim that a complex structure like the eye could not have evolved gradually.

Getting started

Open the simulation and step through each stage of eye evolution one at a time, starting from the simplest light-sensing patch. Pause after each stage before moving to the next.

Tasks

1. Describe the very first stage shown. What can this simple structure detect that a completely flat patch of skin could not?

2. Step to the next stage, where the patch begins to form a cup shape. What new ability does this shape give the organism?

3. For each of the first three stages, briefly explain what survival or reproductive advantage that stage's structure would give an organism, compared to the stage before it.

4. Step forward until a lens-like structure appears. What does a lens add that a simple pinhole opening does not?

5. A common argument against evolution claims a complex eye 'must have appeared all at once' because a half-formed eye would be useless. Using the stages you stepped through, evaluate whether this claim is well supported by the model.

6. Identify one stage in the sequence that you think represents the biggest single jump in complexity. Justify your choice.

7. Explain how natural selection, acting on small variations at each stage, could produce a fully-formed camera-type eye over a very long period of time, without any single 'design' step.

Extension (optional)

Not all animals have camera-type eyes — insects, for example, have compound eyes. Suggest why natural selection might have produced a different eye design in a different lineage, rather than every animal evolving the same 'best' eye.